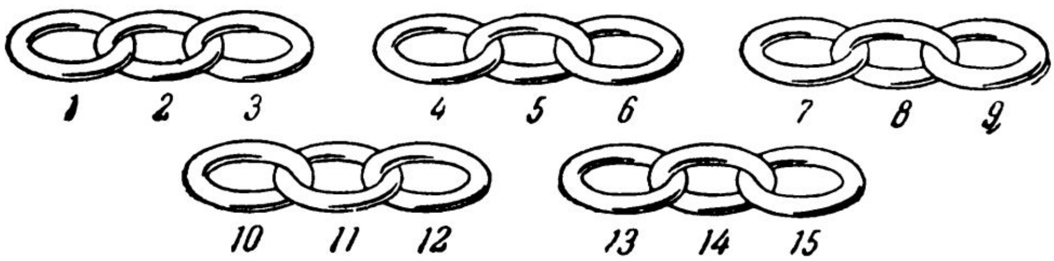
## 13. REPAIRING A CHAIN

Do you know why the young craftsman in the picture is so deep in thought? He has 5 short pieces of chain that must be joined into a long chain. Should he open ring 3




(first operation), link it to ring 4 (second operation), then unfasten ring 6 and link it to ring 7, and so on? He could complete his task in 8 operations, but he wants to do it in 6. How does he do it?

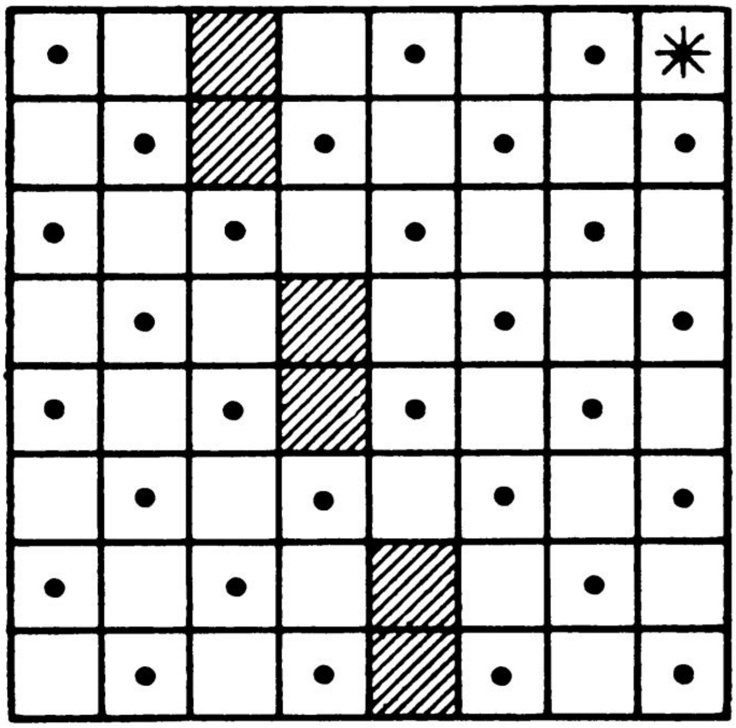
## 11. WOLF, GOAT, AND CABBAGE

A man has to take a wolf, a goat, and some cabbage across a river. His rowboat has enough room for the man plus either the wolf or the goat or the cabbage. If he takes the cabbage with him, the wolf will eat the goat. If he takes the wolf, the goat will eat the cabbage. Only when the man is present are the goat and the cabbage safe from their enemies. All the same, the man carries wolf, goat, and cabbage across the river.

How?

## 6. THE GARDENER'S ROUTE

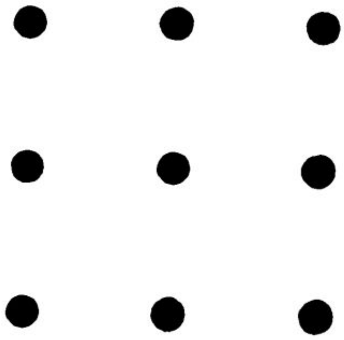
The diagram shows the plan of an apple orchard (each dot is an apple tree). The gardener started with the square containing a star, and he worked his way through



all the squares, with or without apple trees, one after another. He never returned to a square previously occupied. He did not walk diagonally and he did not walk through the six shaded squares (which contain buildings). At the end of his route the gardener found himself on the starred square again.

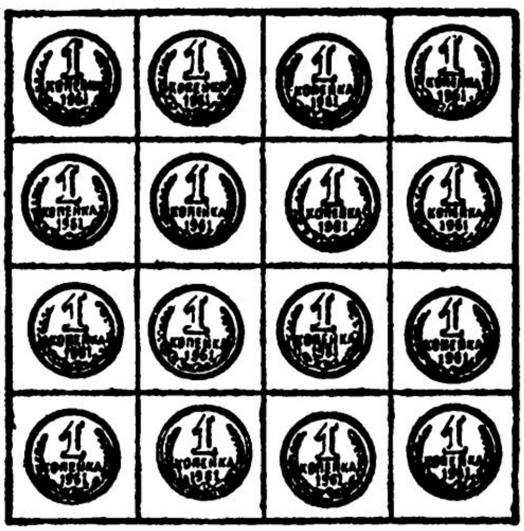
## 24. FOUR STRAIGHT LINES

Make a square with 9 dots as shown. Cross all the dots with 4 straight lines without taking your pencil off the paper.



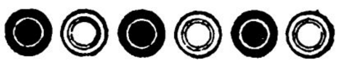
## 21. KEEP IT EVEN

Take 16 objects (pieces of paper, coins, plums, checkers) and put them in four rows of 4 each. Remove 6, leaving an even number of objects in each row and each column. (There are many solutions.)



## 3. MOVING CHECKERS

Place 6 checkers on a table in a row, alternating them black, white, black, white, and so on, as shown.



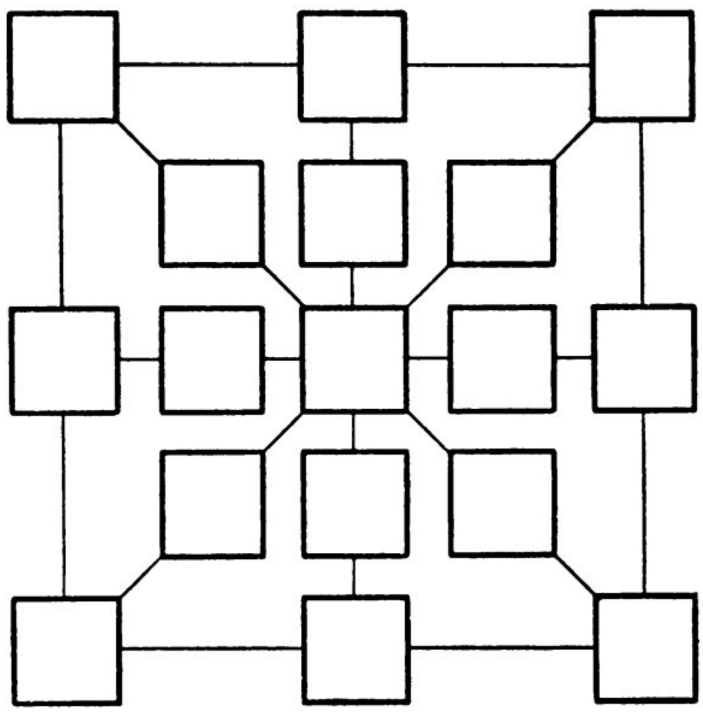
Leave a vacant place large enough for 4 checkers on the left.

Move the checkers so that all the white ones will end on the left, followed by all the black ones. The checkers must be moved in pairs, taking 2 adjacent checkers at a time, without disturbing their order, and sliding them to a vacant place. To solve this problem, only three such moves are necessary.

## 33. PATTERN OF COINS

Take a sheet of paper, copy the diagram on it, enlarging it two or three times, and have ready 17 coins:

| | |
|---|---|
| 20 kopeks | 5 |
| 15 kopeks | 3 |
| 10 kopeks | 3 |
| 5 kopeks | 6 |



Place a coin in each square so that the number of kopeks along each straight line is 55.

[This problem cannot be translated into United States coinage, but you can work on it by writing the kopek values on pieces of paper–*M.G.*]